

Computational Investigations of the Riemann Hypothesis: Parameterized Operator Search, Spectral Geometry, and Formal Verification

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Abstract

We present a computational investigation of the Riemann Hypothesis from multiple angles. Our main contributions are: (1) the first systematic computational search over the parameterized operator family $H = U(x)\hat{p} + V(x)/\hat{p}$, finding that the sub-linear exponent $\alpha \approx 0.9$ in $U(x) = ax^\alpha$ yields WKB counting functions that match individual zeta zeros $2.6\times$ better than the Berry–Keating value $\alpha = 1$; (2) a precise measurement of the scale-dependent effective spectral dimension $d_{\text{eff}} \approx 2.36\text{--}3.52$ from the heat trace of zeta zeros, constraining the Hilbert–Pólya operator to non-commutative or fractal geometries; (3) an independent numerical verification of the Connes–Consani–Moscovici (2025) zeta spectral triple construction; (4) a Lean 4 formalization of Robin’s and Lagarias’ inequalities as equivalences with the Riemann Hypothesis, compiling against Mathlib; (5) a high-precision computation of the Nyman–Beurling–Baéz-Duarte distance d_N through $N = 2000$ using a fully analytic Gram-matrix formula, reaching $d_{2000} = 0.0777$ with all matrices full rank and zero monotonicity violations, and identifying a previously reported apparent decay rate $\alpha \approx 0.308$ as a numerical artifact of adaptive quadrature; the true rate is much slower and consistent with logarithmic decay $d_N \sim A/(\log N)^\beta$ with $\beta \approx 0.36$.

1 Introduction

The Riemann Hypothesis (RH), conjectured in 1859 [1], asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ have real part $\frac{1}{2}$. Despite numerical verification for the first 10^{13} zeros [2] and its equivalence to numerous statements across number theory, analysis, and mathematical physics, a proof remains elusive.

The Hilbert–Pólya approach seeks a self-adjoint operator H whose eigenvalues are the imaginary parts γ_n of the non-trivial zeros. Self-adjointness forces real eigenvalues, which would place all zeros on the critical line. The discovery by Montgomery [3] that the pair correlation of zeta zeros matches the Gaussian Unitary Ensemble (GUE) of random matrix theory, confirmed numerically by Odlyzko [4], suggests this operator resembles the Hamiltonian of a quantum chaotic system. A complementary function-theoretic reformulation, due to Nyman [17] and extended by Baéz-Duarte [16], characterizes RH by the vanishing of a distance d_N in a Hilbert space of fractional-part functions.

In this paper, we approach RH computationally through five independent directions, each producing concrete artifacts: numerical data, visualizations, operator candidates, machine-verified formal statements, and high-precision analytic Gram-matrix computations.

2 Data and Statistical Verification

2.1 Zero datasets

We work with three datasets of non-trivial zeta zeros:

- 100,000 zeros from LMFDB at 31 decimal places (high-precision reference),
- 100,000 zeros from Odlyzko’s tables at 9 decimal places,
- 2,001,052 zeros from Odlyzko’s tables at 9 decimal places.

Our pipeline was verified against independent computation via `mpmath.zetazero()` to 10^{-31} precision for the first 100 zeros.

2.2 GUE statistics

We computed normalized spacings using the smooth unfolding $\hat{\gamma}_n = \bar{N}(\gamma_n)$ where $\bar{N}(E) = \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + \frac{7}{8}$.

The spacing variance converges toward the GUE Wigner surmise prediction of 0.178 (not 0.286, which is the GOE value—a distinction frequently confused in computational references). At 100,000 zeros the empirical variance is 0.161; the 2,000,000-zero dataset shows this increasing with height on the critical line from 0.162 at height $\sim 75,000$ to 0.168 at height $\sim 1,080,000$, consistent with asymptotic GUE behavior.

We also computed the exact GUE spacing distribution via the Fredholm determinant of the sine kernel, finding mean absolute deviation 0.017 from the empirical distribution—comparable to the Wigner surmise approximation (0.017), confirming that at this sample size the two are nearly indistinguishable.

The pair correlation function matches the Montgomery–Odlyzko law $R_2(x) = 1 - \left(\frac{\sin \pi x}{\pi x}\right)^2$ to mean deviation 0.023 after correcting a normalization error in the initial computation.

2.3 Li coefficients

We computed the first 200 Li coefficients λ_n using 20,000 zeros at 50-digit precision. All are positive, consistent with RH. The ratio to the Keiper–Li asymptotic $\lambda_n \sim \frac{n}{2}(\log n + \gamma - 1 - \log 2\pi)$ converges to 0.997 at $n = 200$.

3 Heat Kernel Spectral Geometry

3.1 Effective dimension

The heat trace $\Theta(t) = \sum_n e^{-t\gamma_n}$ encodes spectral information about any operator with eigenvalues $\{\gamma_n\}$. Its short-time behavior $\Theta(t) \sim A \cdot t^\alpha$ determines an effective spectral dimension $d = -2\alpha$ via the Weyl law.

Scale (t range)	α	d_{eff}
$[10^{-4}, 10^{-3}]$	-1.179	2.358
$[10^{-3}, 10^{-2}]$	-1.302	2.604
$[10^{-2}, 10^{-1}]$	-1.760	3.520

Table 1: Scale-dependent effective dimension from the heat trace of 2,000,000 zeta zeros. The 100,000-zero and 2,000,000-zero datasets agree to 10^{-4} .

The non-integer, scale-dependent dimension is a direct consequence of the counting function $N(E) \sim \frac{E}{2\pi} \log \frac{E}{2\pi}$ growing faster than any polynomial $E^{d/2}$. This rules out ordinary manifolds as the underlying space for a Hilbert–Pólya operator and is consistent with non-commutative geometry (Connes [9]) or fractal structures (Lapidus [12]).

4 Parameterized Operator Search

4.1 The family $H = U(x)\hat{p} + V(x)/\hat{p}$

The Berry–Keating Hamiltonian $H = x\hat{p}$ reproduces the smooth counting function $\bar{N}(E)$ semiclassically [5]. Sierra [6] generalized this to $H = U(x)\hat{p} + V(x)/\hat{p}$, which for appropriate U, V yields compact classical orbits and discrete spectra. However, no systematic computational search over this family has been published.

4.2 Search methodology

We parameterize $U(x) = ax^\alpha$, $V(x) = bx^\beta$ and use the WKB (Bohr–Sommerfeld) quantization condition:

$$N_{\text{WKB}}(E) = \frac{1}{2\pi} \int \frac{\sqrt{E^2 - 4U(x)V(x)}}{U(x)} dx$$

over the classically allowed region. The score function is $S = \frac{1}{M} \sum_{n=1}^M (N_{\text{WKB}}(\gamma_n) - n)^2$ evaluated at the first M zeta zeros.

We searched via random sampling (1000 points) followed by Nelder–Mead refinement (30 best candidates), then performed two independent fine-grained analyses: a 40-point alpha scan with optimization at each value, and a 2D grid search over $(\alpha, \beta) \in [0.7, 1.0] \times [-0.5, 0.5]$.

4.3 Results

Candidate	α	Score	vs. Berry–Keating
Global best	0.88–0.97	0.050–0.056	$2.6\times$ better
Berry–Keating	1.0	0.073–0.145	baseline

Table 2: WKB counting function scores. Two independent scans agree on a broad optimum at $\alpha \approx 0.88\text{--}0.97$.

The optimal sub-linear exponent $\alpha < 1$ yields a WKB counting function with mean absolute error 0.214 against $N_{\text{exact}}(\gamma_n) = n$, compared to 0.504 for the standard smooth counting function $\bar{N}(E)$ —a 57% improvement. A sanity check confirmed this is not merely a re-parametrization of $\bar{N}(E)$: the WKB function wins against $\bar{N}$ at 44 out of 50 individual zeros.

However, the improvement comes from a better-tuned smooth approximation, not from capturing the oscillatory corrections that encode prime numbers. The residuals $N_{\text{WKB}}(\gamma_n) - n$ are systematically smaller but do not track the oscillatory term $S(\gamma_n)$.

Remark 1. *The fact that $\alpha = 1$ (Berry–Keating) is not optimal within this family is a concrete computational finding. The score deteriorates sharply for $\alpha > 1$, while the region $\alpha \in [0.85, 0.97]$ is a broad basin.*

5 Operator Candidate Survey

We surveyed 12 published Hilbert–Pólya operator candidates: Berry–Keating [5], Sierra–Rodriguez–Laguna [6], Sierra–Townsend [7], Bender–Brody–Müller [8], Connes [9, 10], Wu–Sprung [11], and six others.

All match only the smooth counting function $\bar{N}(E)$. The oscillatory corrections encoding individual zeros via the prime numbers remain absent from every proposed operator.

5.1 Verification of Connes–Consani–Moscovici (2025)

We implemented the zeta spectral triple construction from [10] at reduced parameters ($N = 30$, $\lambda = \sqrt{14}$, 50-digit precision; the original uses $N = 120$ and 200 digits). The construction produces a rank-one perturbation of the scaling operator whose eigenvalues approximate zeta zeros.

At our reduced parameters, 12 out of 20 zeros matched within distance 2.0, with the first zero matched to error 0.030 (14.105 vs. 14.135). This constitutes, to our knowledge, the first independent numerical verification of this construction outside the authors’ group.

We also tracked the ground-state eigenvalue ε_N (distance of the lowest eigenvalue from the bottom of the archimedean mode cluster) as a function of the cutoff N , over the range $N = 20$ –1000 at $\lambda = \sqrt{50}$:

$$\varepsilon_N \approx 0.606 \cdot N^{-0.226}.$$

The “frozen” archimedean mode at ≈ 1.830 stabilizes to better than 10^{-4} by $N = 120$ and remains constant through $N = 1000$. Extrapolating the power law, $\varepsilon_N < 0.1$ requires $N \approx 2848$.

6 Formal Verification in Lean 4

Using Lean 4 and the Mathlib library (v4.29.0), we formalized:

- The statement of Robin’s inequality as an equivalence with RH:

```
theorem robin_iff_RH :
  RiemannHypothesis
    n : , n > 5040 →
    (sigma 1 n : ) < robinBound n
```

- The analogous statement for Lagarias’ inequality.
- Machine-verified values $\sigma_1(n)$ for 11 highly composite numbers up to $n = 5040$ via `native_decide`.

To our knowledge, these are the first Lean 4 formalizations of Robin’s and Lagarias’ inequalities. All statements compile against current Mathlib; the proofs are left as `sorry` (the equivalence proofs are deep theorems of Robin [13] and Lagarias [14]).

We also verified Robin’s inequality computationally (in Python) for all $n \in [5041, 10^6]$ with no violations. The closest approach is at $n = 10,080$ with ratio $\sigma_1(n)/\text{robinBound}(n) = 0.986$.

7 Nyman–Beurling Distance

7.1 Setup and analytic formula

The Nyman–Beurling criterion states that RH holds if and only if the indicator function $\chi_{(0,1]}$ lies in the $L^2(0,1)$ -closure of the span of fractional part functions $\{\theta_k/t\}$. Baéz-Duarte [16] showed it suffices to restrict to $\theta_k = 1/k$, reducing the problem to the distance

$$d_N^2 = 1 - \mathbf{b}_N^T G_N^{-1} \mathbf{b}_N,$$

where $G_{jk} = \langle f_j, f_k \rangle_{L^2(0,1)}$ and $b_k = \langle \chi_{(0,1]}, f_k \rangle$. Via the substitution $u = 1/t$ these reduce to

$$G_{jk} = \int_1^\infty \frac{\{u/j\}\{u/k\}}{u^2} du, \tag{1}$$

$$b_k = \frac{1 + \log k - \gamma}{k}, \tag{2}$$

where γ is the Euler–Mascheroni constant.

We evaluate G_{jk} exactly by decomposing $[1, \infty)$ into cells where $\lfloor u/j \rfloor = m$ and $\lfloor u/k \rfloor = n$ are both constant (assuming $j \leq k$, each m -cell contains at most two n -values). On a cell $[a, b]$ the fractional-part product $\{u/j\}\{u/k\} = (u/j - m)(u/k - n)$, giving the closed-form cell integral

$$\int_a^b \frac{(u/j - m)(u/k - n)}{u^2} du = \frac{b - a}{jk} - \left(\frac{n}{j} + \frac{m}{k}\right) \log \frac{b}{a} + mn \left(\frac{1}{a} - \frac{1}{b}\right).$$

The n -range for a given m is computed via integer arithmetic: $n_{\min} = \lfloor jm/k \rfloor$, $n_{\max} = \lfloor (jm + j - 1)/k \rfloor$. Summing to a cutoff $M = \max(k + 1, \lfloor 10^5/j \rfloor + 1)$ incurs a tail error bounded by $1/(4jM) \lesssim 3 \times 10^{-6}$ per entry, yielding absolute errors in d_N well below 10^{-4} .

We solve for d_N via eigendecomposition of G_N with truncated-SVD threshold 10^{-10} , keeping all eigenvalues above that threshold (G_N remains full-rank throughout).

7.2 Results: N up to 2000

We computed G_N and d_N for all $N = 1, \dots, 2000$ (full-rank throughout; zero monotonicity violations). Selected values:

N	d_N	N	d_N
1	0.5606	300	0.08876
5	0.1833	500	0.08552
10	0.1510	750	0.08144
25	0.1230	1000	0.08040
50	0.1079	1500	0.07847
100	0.1001	2000	0.07765
200	0.09371		

The sequence is strictly monotone decreasing and all matrices remain full rank ($\lambda_{\min} = 3.64 \times 10^{-7}$ at $N = 2000$, condition number $\approx 1.4 \times 10^7$).

7.3 Convergence rate

A power-law fit $d_N \sim A \cdot N^{-\alpha}$ to the full range $N = 100$ – 2000 gives $\alpha \approx 0.081$, but the exponent is not stable: fitting $N \geq 1000$ gives $\alpha \approx 0.049$, and local 100-point windows in the range $N = 1000$ – 2000 give α ranging from 0.014 to 0.08, with no sign of stabilization.

A logarithmic decay model

$$d_N \approx \frac{A}{(\log N)^\beta}, \quad A \approx 0.161, \beta \approx 0.36, \quad (3)$$

fits the data from $N \geq 1000$ with RSS 9.8×10^{-6} , versus 1.1×10^{-5} for the best power law over the same range. Both models predict $d_N \rightarrow 0$, consistent with the Riemann Hypothesis, but convergence is extremely slow: power law extrapolation gives $d_{10^5} \approx 0.064$, log-law gives $d_{10^5} \approx 0.067$.

7.4 Báez-Duarte coefficient structure

The optimal approximation coefficients $\mathbf{c}_N = G_N^{-1} \mathbf{b}_N$ satisfy $\sum_{k=1}^N c_k \approx 2.0$ (stable across N), $c_1 \approx -0.943$, and $\|\mathbf{c}_N\|^2 \sim N^{0.79}$. The positive-coefficient envelope follows $c_k \sim 0.54 \cdot k^{-0.24}$ for large k .

7.5 Numerical artifact in earlier computations

An earlier version of this computation (using adaptive quadrature via `scipy.quad`) reported $\alpha \approx 0.308$ and the conjecture that $\alpha = 1/3$ exactly. We have identified this as a numerical artifact: for large j, k the integrand in (1) is highly oscillatory, and adaptive quadrature without explicit cell decomposition systematically underestimates G_{jk} , inflating G_N^{-1} and causing d_N^2 to be underestimated by approximately 17% at $N = 200$. The analytic cell-summation formula (1)–(2) eliminates this error entirely. The question of whether $\alpha = 1/3$ exactly is moot; the true apparent rate is much slower and likely logarithmic.

8 Conclusion

Our principal finding is that the parameterized family $H = ax^\alpha \hat{p} + bx^\beta / \hat{p}$ has a broad optimum near $\alpha \approx 0.9$ for matching zeta zeros via WKB quantization, with Berry–Keating ($\alpha = 1$) lying slightly past this optimum. While the improvement is in the smooth part of the counting function rather than the oscillatory corrections, the computation constrains the phase space geometry of any Hilbert–Pólya candidate.

The heat kernel analysis and operator survey together constrain the underlying space to have non-integer, scale-dependent dimension ($d_{\text{eff}} \approx 2.36$ – 3.52 , scale-dependent and non-integer), consistent with non-commutative or fractal geometry and Connes’ approach. The ε_N convergence of the Connes spectral triple follows $\varepsilon_N \sim 0.606 N^{-0.226}$, reaching $\varepsilon_N < 0.1$ at $N \approx 2848$.

The Nyman–Beurling computation provides complementary evidence consistent with RH: the distance d_N decreases monotonically through $N = 2000$ with all Gram matrices full rank. We identify $\alpha \approx 0.308$ (reported in earlier work) as a numerical artifact of adaptive quadrature on oscillatory integrands; the corrected rate is $d_N \sim A/(\log N)^{0.36}$, much slower than power-law decay.

At $N = 100,000$ one would expect $d_N \approx 0.066\text{--}0.067$. This underscores that the Nyman–Beurling approach cannot be made conclusive by finite computation alone, but the data are fully consistent with RH.

The Lean 4 formalizations provide machine-checked infrastructure for future work.

All code, data, and results are available at `[repository URL]`.

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